

ML COMPETITION PLATFORM

Complete Guide: Setting Up Competitions & Understanding Scoring

5

Steps to Launch

3

Score Components

92%

Max Score Cap

3/day

Submission Limit

What This Guide Covers

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| 1 | Setting up your competition in the Admin Panel — title, rules, prizes, deadlines |
| 2 | Creating the database: all SQL tables needed for competitions and submissions |
| 3 | Inserting ground truth labels using SQL — no CSV upload needed |
| 4 | How the scoring engine calculates Accuracy, F1, and Code Mastery scores |
| 5 | The final score formula and how the leaderboard ranks participants |

STEP 1

Create Competition in Admin Panel

Log into your Admin Dashboard and fill in the competition form. Every field below affects how the competition runs and how scores are calculated.

Field	Example Value	Purpose
Title	Kigali Crop Disease Challenge	Shown on leaderboard and competition list
Summary	Classify maize leaf images into 5 classes	Short description shown in cards
Description	Full problem statement with dataset details	Full brief shown to participants
Rules	One rule per line	Displayed in the rules tab
Prize	RWF 100,000	Shown as motivation
Tags	Computer Vision, Agriculture	Used for filtering
Deadline	2 weeks from today	Closes submissions after this date
Max Submissions/Day	3	Enforced per user per day
Status	open	Set to open when ready to accept submissions
Benchmark Score	45.0	Baseline score shown on leaderboard

After saving: Go to Supabase Table Editor → competitions table → copy the UUID from the id column. You will need it for Step 3.

STEP 2

Create Database Tables

Run this SQL once in your Supabase SQL Editor to create all required tables:

```
-- COMPETITIONS TABLE CREATE TABLE IF NOT EXISTS competitions ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), title TEXT NOT NULL, summary TEXT, description TEXT, rules TEXT, prize TEXT, tags TEXT[], deadline TIMESTAMPTZ, status TEXT DEFAULT 'open', max_submissions_per_day INT DEFAULT 3, benchmark_score FLOAT, created_at TIMESTAMPTZ DEFAULT now() ); -- GROUND TRUTH TABLE (hidden from participants) CREATE TABLE IF NOT EXISTS ground_truth ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), competition_id UUID REFERENCES competitions(id), row_id TEXT NOT NULL, true_label TEXT NOT NULL, difficulty_tier TEXT DEFAULT 'medium', adversarial BOOLEAN DEFAULT false, column_weight FLOAT DEFAULT 1.0 ); -- SUBMISSIONS TABLE CREATE TABLE IF NOT EXISTS submissions ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), competition_id UUID REFERENCES competitions(id), username TEXT NOT NULL, model_name TEXT, accuracy_score FLOAT, f1_score FLOAT, code_score FLOAT, final_score FLOAT, feedback JSONB, created_at TIMESTAMPTZ DEFAULT now() );
```

RLS Note: Make sure Row Level Security is configured so the frontend can read submissions and competitions. Run: ALTER TABLE submissions DISABLE ROW LEVEL SECURITY; if you face empty leaderboard issues.

STEP 3

Insert Ground Truth via SQL

Ground truth is the secret answer key. Only the admin inserts it — participants never see it. Each row has a difficulty tier and weight that affect scoring.

Tier	Weight	Meaning	Example
easy	1.0x	Straightforward prediction	Clear healthy leaf
medium	1.5x	Requires feature engineering	Early stage disease
hard	2.0x	Subtle or ambiguous class	Rare disease variant

Ground Truth SQL Template:

```
-- Replace 'YOUR-COMPETITION-UUID' with the real UUID from Step 1
INSERT INTO ground_truth
(competition_id, row_id, true_label, difficulty_tier, adversarial, column_weight) VALUES
('YOUR-COMPETITION-UUID', '1', 'healthy', 'easy', false, 1.0), ('YOUR-COMPETITION-UUID', '2',
'common_rust', 'medium', false, 1.5), ('YOUR-COMPETITION-UUID', '3', 'gray_leaf_spot', 'hard',
false, 2.0), ('YOUR-COMPETITION-UUID', '4', 'streak_virus', 'hard', false, 2.0), --
adversarial=true rows are hidden bonus rows: ('YOUR-COMPETITION-UUID', '5', 'healthy', 'easy',
true, 1.0);
```

Adversarial rows: Mark adversarial=true for trick rows. Correctly predicting them gives a +2.0 bonus to the final score. Participants do not know which rows are adversarial.

STEP 4

Scoring Engine — How Scores Are Calculated

When a participant uploads their CSV, the scoring engine runs 3 calculations then combines them into one final score.

ACCURACY SCORE	Compares predicted_label vs true_label for every non-adversarial row. Each correct prediction earns its difficulty weight. Hard rows are worth 2x easy rows.
MACRO F1 SCORE	Calculates Precision and Recall for each class separately, then averages them. Prevents gaming by always predicting the most common class.
CODE MASTERY	Checks how many of the 8 required columns are present in the submission CSV. Rewards participants who document their approach properly.

STEP 5

Final Score Formula

Score Cap: The maximum possible score is 92%. This is by design — it prevents any single team from achieving a "perfect" score and keeps the competition meaningful. The adversarial bonus can push scores above the raw composite but never past 92%.

Worked Example

Component	Raw Score	Weight	Contribution
Accuracy	80.0%	x 0.45	36.0%
Macro F1	75.0%	x 0.30	22.5%
Code Mastery	100.0%	x 0.25	25.0%
Adversarial Bonus	+2.0	—	+2.0%
FINAL SCORE			85.5% (capped at 92%)

Required Submission CSV Columns

Participants must include these columns in their CSV. Missing columns reduce the Code Mastery score.

Column	Required?	Description
row_id	YES	Must match the row IDs in the test set
predicted_label	YES	The class prediction for each row
confidence	Code	Model confidence score (0.0 to 1.0)
model_name	Code	Name of the model used
preprocessing_steps	Code	What preprocessing was applied
feature_count	Code	Number of features used
validation_strategy	Code	e.g. kfold, hold-out
used_cross_validation	Code	true or false
handled_class_imbalance	Code	true or false
train_test_split_ratio	Code	e.g. 0.8

How the Leaderboard Works

Ranked by final_score

Highest score appears at rank #1. Medals shown for top 3.

Best submission only

Only each participant's best submission is shown — not all attempts.

Live updates	Leaderboard refreshes automatically after each submission.
Daily limit enforced	Once the daily limit is reached, submissions are blocked until midnight.
YOU badge	Your own row is highlighted so you can find your rank instantly.